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# ВЫПУСКНАЯ КВАЛИФИКАЦИОННАЯ РАБОТА
# MASTER THESIS

*Тема* Оценка эффекта международных санкций на оборонные расходы воюющих стран

*Title* Assessing the Impact of International Sanctions on Military Spending of Warring Countries

*Студент/Student* ____________________ Кривенко Глеб Иванович / Gleb Krivenko
Подпись/Signature

*Руководитель/Advisor* ____________________ Петрова Мария Алексеевна / Maria Petrova
Подпись/Signature
PhD, Профессор РЭШ / Professor, NES

*Оценка/Grade*
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**Abstract**

This thesis investigates the causal impact of international economic sanctions on the military expenditure of target states. Assessing this relationship is fundamentally complicated by endogeneity, as sanctions are disproportionately imposed on regimes already undergoing rapid militarization. Utilizing a country-year panel dataset from 1949 to 2023, this study employs a 2SLS identification strategy to overcome selection bias. As a primary methodological contribution, the study constructs exogenous sanction intensity measures by weighting bilateral trade restrictions using predicted trade shares derived from a structural gravity model.

The empirical results reveal that the positive correlation between sanctions and military spending observed in standard OLS models is entirely driven by reverse causality; aggregate sanctions exert no statistically significant causal effect during peacetime. However, disaggregating by sanction type and conflict status reveals substantial heterogeneity. While arms embargoes and financial restrictions show no robust impact, targeted military assistance sanctions and partial export restrictions significantly reduce absolute defense spending, but exclusively when the target is engaged in active armed conflict. Conversely, comprehensive trade embargoes trigger a "siege economy" response, prompting regimes to protect or increase the military's share of a shrinking GDP. The findings suggest that coercive statecraft is most effective when creating physical logistical bottlenecks during wartime, and that sanctioning coalitions should be formed based on structural, gravity-driven trade dependencies rather than aggregate economic weight.

# 1 Introduction

Economic sanctions have emerged as a primary instrument of modern coercive statecraft, frequently deployed as an alternative to military intervention. A fundamental objective of many sanction regimes is to constrain the target state's capacity to wage war or project power by degrading its military-industrial base. However, the empirical evaluation of whether international coercion successfully reduces a target's military expenditure remains highly contested. While some theories suggest that economic isolation forces regimes to cut defense budgets due to resource scarcity, competing frameworks argue that external pressure triggers a "rally-around-the-flag" effect, prompting targeted governments to consolidate power and paradoxically increase military outlays as a defensive measure.

Evaluating these competing hypotheses is severely complicated by endogeneity. The imposition of sanctions is a non-random, strategic decision: sender states typically target regimes that are already exhibiting aggressive foreign policies or undergoing rapid internal militarization. Consequently, naive empirical estimations (such as standard OLS regressions) consistently show a positive correlation between sanction pressure and defense spending. The methodological challenge lies in determining whether this positive relationship is a causal reaction to the sanctions themselves, or merely a manifestation of selection bias and reverse causality. Furthermore, existing cross-country studies frequently treat sanctions as uniform, binary shocks, failing to account for the specific type of restriction (e.g., arms embargoes versus financial freezes) and the actual economic vulnerability of the target to the sanctioning coalition.

This study aims to isolate the causal impact of international sanctions on the military expenditure of target states. Utilizing a country-year panel dataset from 1949 to 2023, we estimate the effect of sanctions on both the absolute volume of defense spending and the military burden (military expenditure as a share of GDP). To overcome the fundamental endogeneity problem, we employ a 2SLS identification strategy. Following recent advancements in the empirical literature, the instrument is constructed via a zero-stage dyadic model that predicts the probability of sanction imposition using strictly exogenous predictors, namely the lagged political distance between states in the UN General Assembly and the historical sanctioning aggressiveness of the sender.

To capture the true economic exposure of the target, this study introduces a methodological innovation by weighting trade sanctions using bilateral trade shares predicted by a structural gravity model. This gravity-based IV approach effectively purges the measurement of sanction intensity from the contemporaneous political shocks and trade distortions that typically confound such analyses.

The findings of this paper contradict the initial correlational evidence: once selection bias is accounted for, aggregate sanctions demonstrate no statistically significant causal effect on military expenditure during peacetime. The positive correlation observed in standard OLS models is entirely driven by endogeneity; sanctions do not compel states to arm, but are rather imposed on states that are already arming.

However, disaggregating the analysis by sanction type and conflict status reveals substantial heterogeneity. While arms embargoes and financial restrictions show no robust effect, targeted military assistance sanctions and partial export restrictions (specifically, the denial of export markets to the target) induce a severe and causal reduction in absolute military spending, but only when the target is engaged in an active armed conflict. This suggests that sanctions are effective primarily when they create physical and financial bottlenecks in a war economy that is already operating at capacity. Conversely, we find evidence that comprehensive export embargoes can provoke a "siege economy" response during wartime, where the targeted regime protects the military's share of a shrinking national budget, despite facing severe economic contraction.

By demonstrating the failure of aggregate, unweighted sanction measures and highlighting the specific conditions under which targeted coercion succeeds, this study contributes to the literature on economic statecraft and provides a deeper framework for evaluating the fiscal consequences of international sanctions.

## 2 Background

The central question regarding the imposition of international sanctions is whether such measures can effectively compel a target government to alter its behavior. This decision involves a calculated trade-off: the sender must weigh the probability of inducing policy change against the self-inflicted economic costs, which often manifest as reduced trade turnover, disrupted value chains, and diminished output. Effectively, sanctioning states must determine their willingness to pay for each unit of economic damage inflicted upon the target (De Souza et al., 2024).

Beyond the direct economic calculus, the literature highlights significant political risks associated with coercive statecraft. A primary concern is that external pressure may inadvertently strengthen an authoritarian regime. Economic hardship often triggers a "rally-around-the-flag" effect, which ruling elites can exploit to justify the consolidation of power and the suppression of domestic dissent (Gershenson, 2002). The intensity of this effect is frequently conditional on the target's initial institutional characteristics, including its degree of democratization and the inherent stability of the regime (Dizaji and Farzanegan, 2024). Furthermore, broad-based sanctions often lead to a measurable decline in the quality of life for the general population, occasionally precipitating humanitarian crises characterized by increased poverty and systemic violence (Peksen and Drury, 2010; Naylor, 2001). These humanitarian externalities often force policymakers to moderate the severity of restrictive measures, even when more aggressive interventions might appear more instrumentally effective.

The efficacy of sanctions in de-escalating or preventing armed conflict has become a particularly salient issue in recent years. The restrictive regimes imposed on Russia following the 2022 invasion of Ukraine (van Bergeijk, 2022), as well as long-standing pressure on states such as Iran and North Korea, have brought this debate to the forefront. These cases have largely demonstrated the limits of economic coercion; in many instances, attempts to secure a settlement in exchange for sanctions relief have failed. Evidence from the latest release of the Global Sanctions Data Base suggests that the coercive impact of sanctions can be significantly neutralized by trade rerouting toward non-sanctioning partners. Specifically, Yalcin et al. (2025) indicate that reduced trade costs between Russia and major economies such as China, India, and Turkey may effectively offset Western restrictions, potentially resulting in a net benefit for the target state rather than the intended economic exhaustion. Such persistent ineffectiveness necessitates a critical reassessment of sanctions as a tool of international relations and challenges the assumption that economic pain automatically translates into diplomatic concessions (Peterson, 2023).

Among the various instruments of coercion, trade restrictions remain the most prominent. These measures range from targeted bans on specific product groups — such as dual-use technologies, sensitive raw materials, or, relevant to this study, armaments — to comprehensive embargoes on all commercial exchange. In some cases, the enforcement of these measures is augmented by secondary sanctions targeting third-party entities that continue to engage with the recipient. Trade sanctions are often regarded as the most potent tool in the sender's arsenal due to the deep integration of modern value chains and the global division of labor. The disruptive potential of these measures is highest when bilateral trade flows are substantial or when the recipient state exhibits a structural dependence on the sender, as is often the case within integrated economic unions (Escriba-Folch, 2010).

Beyond trade, coercive measures frequently target capital flows and broader financial transactions. These interventions are designed to exploit specific financial vulnerabilities: they may aim at economies heavily reliant on foreign direct investment or, conversely, target capital-rich states by freezing external sovereign reserves. In either case, financial restrictions increase the costs and technical complexity of international settlements, thereby creating significant friction for all forms of cross-border commercial activity.

More targeted interventions specifically address the defense sector through the suspension of military cooperation and the imposition of arms embargoes. While these measures are a logically consistent tool for curbing a state's war-making capacity, they are often implemented during the nascent stages of a diplomatic crisis. Consequently, such baseline restrictions are frequently already in force by the time a state transitions to active hostilities, which may limit their marginal utility as a deterrent against openly bellicose regimes.

A final category of sanctions includes measures that are often perceived as less economically damaging, serving either an expressive or an isolationist function. Diplomatic sanctions, for instance, may represent a purely formal signal of international disapproval. More restrictive isolationist policies — such as the suspension of air, maritime, or terrestrial transport and the tightening of visa regimes — aim to sever physical links between the sender and the target. While the direct economic impact of such measures is often secondary, they are frequently justified on the grounds of national security, aiming to mitigate risks related to espionage or subversion. Although the effectiveness of these security-driven policies is a critical aspect of statecraft, its assessment lies beyond the empirical scope of this study.

Methodologically, the existing literature on the economic consequences of sanctions is characterized by a prevalence of idiosyncratic case studies. Researchers frequently employ synthetic Difference-in-Differences or synthetic control methods to quantify the impact of specific restrictive regimes on individual national economies (Farzanegan, 2022; Mayberry, 2023). While these approaches offer high internal validity and precise policy insights for a given country, their findings are inherently difficult to generalize. A multi-country synthetic DiD framework could theoretically provide a more robust basis for a generalizable "forecast" of how sanctions affect development and military outlays; however, such comparative large-N studies remain limitedly represented in contemporary research.

Despite the dominance of case studies, a growing body of comparative research has begun to address these dynamics within a cross-country framework. For instance, (Escriba-Folch, 2012) examines the budgetary responses of different regime types to international pressure, finding that authoritarian leaders tend to redistribute resources toward their core institutional beneficiaries. Under sanction-induced stress, military regimes frequently increase defense outlays to ensure elite loyalty, whereas personalist regimes may reduce social spending in favor of increased internal repression to stifle dissent. While providing essential theoretical foundations, such large-N studies often treat sanctions as a binary or undifferentiated phenomenon. This lack of granularity regarding the severity and scope of restrictive measures potentially masks significant variation in how target states respond to different levels of coercion.

Consequently, a critical gap remains in the literature: the challenge of systematically comparing not only the stringency of sanctions but also the relative importance of the sanctioning entities. While certain studies have attempted to weight the intensity of sanctions (Dizaji and Farzanegan, 2019), these efforts have been largely confined to single-country analyses where the specifics of each policy shift are more easily quantified. In a broader cross-country panel, however, a standardized method for assessing the aggregate "strength" of heterogeneous sanctioning coalitions has remained elusive.

This study proposes to address this methodological deficit by developing an aggregation framework that accounts for both the qualitative severity of the measures and the structural significance of the sender states. By differentiating between partial and complete restrictions and weighting these measures by the economic and trade-based relevance of the sanctioning coalition, we aim to provide a more precise estimation of the causal pressure exerted on a target's military budget. This approach allows for an empirical test of whether the defense-sector responses observed in previous literature are driven by the mere presence of international disapproval or by the objective economic cost imposed by the sender's global standing.

Drawing on the theoretical debates regarding coercion, regime survival, and resource allocation discussed above, three primary hypotheses are formulated in this study to be tested empirically.

The existing literature often posits a direct link between external pressure and increased defense spending, driven by a regime's need to consolidate power or signal resolve (the "rally-around-the-flag" effect). However, we argue that this relationship is fundamentally confounded by selection bias: sanctions are disproportionately levied against states already pursuing aggressive or militaristic policies. Once this endogeneity is appropriately addressed, we expect the apparent positive impact of broad sanctions to dissipate. Thus, our first hypothesis assumes a null baseline effect:

**Hypothesis 1:** *Once corrected for endogeneity and selection bias, aggregate international sanctions exert no significant causal effect on the military expenditure of target states in peacetime.*

While broad economic restrictions may fail to alter budgetary priorities, highly targeted measures directly affecting the defense sector's supply chains should yield different outcomes. This is particularly relevant during active armed conflicts, where the rapid consumption of materiel makes the target state highly sensitive to logistical disruptions. we anticipate that constraints on military aid and components will create insurmountable physical bottlenecks.

**Hypothesis 2:** *Military assistance sanctions causally reduce absolute military expenditure, but this effect is heterogeneous and strictly conditional on the target state being engaged in an active armed conflict.*

Finally, the impact of trade sanctions requires distinguishing between the direction of restricted flows. For the target state, the loss of export markets (imposed via export sanctions by the sender) results in a direct reduction of hard currency revenues. We theorize that the regime's response to such severe economic contraction will depend on the intensity of the threat. Under partial restrictions, the state may be forced to reduce military outlays. However, under comprehensive economic isolation, the regime is likely to perceive an existential threat, prompting it to shield the defense sector from budgetary cuts even as the overall economy shrinks.

**Hypothesis 3:** *The causal impact of trade sanctions depends on their scope. Comprehensive restrictions on the target's exports (sender's imports) induce a "siege economy" response, leading to an increase in the military burden (military expenditure as a share of GDP), whereas partial restrictions tend to diminish it.*

# 3 Data and Variables

To empirically investigate the impact of international sanctions on the military expenditure of target states, we construct a country-year panel dataset. The baseline sample covers the period from 1949 to 2023, encompassing 116 countries. The final dataset is merged from several publicly available sources, with country identifiers standardized to the ISO-3 alpha code format.

## 3.1 Dependent Variables

Our primary focus is on the military burden and the absolute volume of defense resources of the target states. Data on military spending is sourced from the Stockholm International Peace Research Institute (SIPRI) Military Expenditure Database (2024). To ensure the robustness of our findings and to clarify the applicable mechanism of the impact, we employ two alternative dependent variables:

- **Log of Real Military Expenditure ($\ln(MilExp_{it})$)**: The natural logarithm of the target country's military expenditure, expressed in constant 2023 US dollars. This captures the absolute change in the volume of resources allocated to defense.
- **Military Expenditure as a Share of GDP ($MilShare_{it}$)**: The ratio of military spending to the country's Gross Domestic Product. This variable reflects the priority of the defense sector relative to the overall economy, mitigating concerns that changes in absolute spending are merely driven by general economic contraction under sanctions.

## 3.2 Key Independent Variables

Data on international sanctions are extracted from the Global Sanctions Data Base (GSDB) (Syropoulos et al., 2023; Felbermayr et al., 2020; Kirilakha et al., 2021), which provides dyadic (sender-target-year) information on various types of sanctions. we aggregate this dyadic data to the country-year level (target-year) to construct our main explanatory variables. we distinguish between overall sanction pressure and specific types of sanctions: *arms embargoes, military, trade*(which will be further disaggregated into partial and complete restrictions), *financial, travel, and other sanctions.*

To account for the intensity and economic weight of the sanctioning coalition, two distinct aggregation methods were employed:

1. **Log Count of Senders**: The natural logarithm of one plus the number of unique sending countries that imposed sanctions against target $i$ in year $t$ ($\ln(1+\text{Count}_{it})$).

2. **GDP-Weighted Sanction Pressure**: Recognizing that sanctions from major economies impose higher costs, we construct a weighted measure by summing the real GDP (sourced from SIPRI) of all sending countries imposing sanctions on target $i$ in year $t$, and taking the natural logarithm of one plus this sum ($\ln(1 + \sum GDP_{sender,t})$).

To test for heterogeneous effects of sanctions during armed conflicts, we introduce an interaction term between our sanction measures and a binary war indicator ($Sanctions_{it} \times War_{it}$).

## 3.3 Instrumental Variables (The Dyadic Stage)

To address the potential endogeneity of sanctions (e.g., reverse causality or omitted variable bias), an IV strategy will be employed. Following Kwon et al. (2022), we first estimate the probability of sanction imposition at the dyadic level (sender-target-year) using a probit model. The predicted probabilities are then aggregated to serve as an instrument in the second-stage country-year regressions.

The instruments used in the zero-stage probit model rely on exogenous factors influencing the sender's propensity to sanction:

- **Sender aggressiveness**: A "leave-one-out" measure capturing the sender's general propensity to use sanctions. It is calculated as the natural logarithm of one plus the total number of sanctions imposed by the sender globally in year $t$, strictly excluding any sanctions against the specific target $j$. To avoid simultaneity, we use the one-year lagged value.

- **Political distance**: A measure of political misalignment between the sender and the target, derived from the United Nations General Assembly (UNGA) voting records (UN Dag Hammarskjöld Library, 2024). The average absolute difference in voting choices (Yes=1, No=0, Abstain=0.5) between the dyad is calculated; the one-year lagged value will be used to ensure exogeneity.

- **Interaction of aggressiveness and distance**: We interact the lagged sender aggressiveness with the lagged UNGA voting distance to capture non-linearities in the sender's behavior towards politically distant targets.

- **Gravity controls**: we also include standard exogenous geographic and historical dyadic controls sourced from the CEPII GeoDist database (Mayer and Zignago, 2011): the natural log of geographic distance ($\ln(Distance_{ij})$), contiguity, common official language, and colonial links.

## 3.4 Trade Data and Gravity Variables

While simple counts or GDP-based measures capture the aggregate size of the sanctioning coalition, they fail to account for the actual economic vulnerability of the target state. A sanction from a country representing 50% of the target's export market is vastly more impactful than a sanction from a similarly sized economy with zero bilateral trade.

To address this, we construct exposure weights based on bilateral trade flows. Data on international trade is sourced from the Correlates of War (CoW) Project's Bilateral Trade dataset (version 4.0) (Palmer et al., 2015), which provides dyadic import and export flows expressed in current US dollars.

However, because actual trade flows decline mechanically when sanctions are imposed (introducing severe endogeneity), we do not use raw trade data to weight sanctions. Instead, we use the CoW trade data in conjunction with standard gravity model predictors to generate strictly exogenous, predicted trade shares (the detailed methodology for this procedure is discussed in Section 5.3).

To estimate this structural gravity model, we rely on exogenous, time-invariant geographic and historical dyadic frictions sourced from the CEPII GeoDist database (Mayer and Zignago, 2011). These gravity controls include the natural logarithm of the geographic distance between capital cities ($\ln(Distance_{ij})$), a binary indicator for contiguous borders, a dummy for common official language, and an indicator for historical colonial links.

### 3.5 Control Variables

To isolate the effect of sanctions, controls for several confounding factors at the target-country level were introduced:

- **War**: A dummy variable equal to 1 if the target country is involved in a militarized interstate dispute (MID) reaching the level of the "Use of Force" or "War," based on the Correlates of War (CoW) MID dataset (Palmer et al., 2015).
- **Economic Size ($\ln(GDP_{it})$)**: The natural logarithm of the target's real GDP (in constant 2023 US dollars), controlling for the resource base. *Note: This control is omitted when Military Expenditure as a Share of GDP is used as the dependent variable.*
- **Regime Type ($Polity_{it}$ and $Polity_{it}^2$)**: The Polity V score (Marshall and Gurr, 2020), ranging from -10 (strongly autocratic) to +10 (strongly democratic), controls for domestic institutional constraints. Squared term is included to account for potential non-linear (U-shaped) relationships between democratization and military spending.
- **Lagged Dependent Variable**: To account for the strong bureaucratic inertia inherent in defense budgeting, we control for the one-year lag of the dependent variable.

Finally, all our main specifications include **country fixed effects** to absorb unobserved, time-invariant heterogeneity across states (e.g., geographic vulnerability, strategic culture), and **year fixed effects** to control for global shocks affecting all nations simultaneously (e.g., the end of the Cold War, global financial crises).

Summary statistics are shown in Table 1.

Table 1: Summary Statistics

| **Variable** | **Obs.** | **Mean** | **Std. Dev.** | **Min.** | **Max.** |
|---|---|---|---|---|---|
| Log Real Military Expenditure | 5068.0 | 20.49 | 2.446 | 10.728 | 27.621 |
| Military Expenditure (% of GDP) | 5068.0 | 0.029 | 0.034 | 0.0 | 1.173 |
| Sanctions (Count of Senders) | 5068.0 | 10.392 | 34.647 | 0.0 | 193.0 |
| Sanctions (Log Count) | 5068.0 | 0.831 | 1.349 | 0.0 | 5.268 |
| War (Dummy) | 5068.0 | 0.024 | 0.154 | 0.0 | 1.0 |
| Log Real GDP | 5068.0 | 24.398 | 2.334 | 13.455 | 33.698 |
| Polity V Score | 5068.0 | 2.323 | 7.231 | -10.0 | 10.0 |

Table 2: *Notes:* Descriptive statistics are calculated based on the estimation sample used in the baseline regressions. Observations with missing values in key control variables have been excluded.

# 4 Empirical Strategy: Baseline OLS Models

To investigate the relationship between international sanctions and the military expenditure of target states, we begin by estimating a series of baseline OLS models. These models serve to establish the initial correlation between our variables of interest and to demonstrate the importance of controlling for unobserved heterogeneity before proceeding to our main IV strategy.

Our baseline empirical approach relies on panel regressions to exploit both cross-sectional (between-country) and time-series (within-country) variation. The most comprehensive specification of our baseline model takes the following form:

$$Y_{it} = \beta_0 + \beta_1 Sanctions_{it} + \beta_2 War_{it} + \beta_3 (Sanctions_{it} \times War_{it}) + \gamma' \mathbf{X}_{it} + \alpha_i + \tau_t + \varepsilon_{it} \quad (1)$$

Where:

- $Y_{it}$ represents the dependent variable for country $i$ in year $t$. Depending on the specification, this is either the natural logarithm of real military expenditure ($\ln(MilExp_{it})$) or military expenditure as a share of GDP ($MilShare_{it}$).

- $Sanctions_{it}$ denotes the measure of sanction pressure imposed on country $i$ in year $t$. we utilize two alternative operationalizations: the log-transformed count of sanctioning states ($\ln(1 + Count_{it})$) and the log-transformed sum of the real GDP of all sanctioning states ($\ln(1 + \sum GDP_{sender,t})$). We estimate Equation 1 separately for aggregate sanctions and for specific sanction types.

- $War_{it}$ is a binary indicator equalling 1 if the target country is involved in a militarized interstate dispute (MID) reaching the level of the "Use of Force" or "War", and 0 otherwise. This acts as a main effect, capturing the direct impact of armed conflict on military spending.

- $(Sanctions_{it} \times War_{it})$ is the interaction term between the sanction measure and the war indicator. This term allows us to test whether the marginal effect of sanctions differs systematically during wartime compared to peacetime.

- $\mathbf{X}_{it}$ is a vector of additional time-varying control variables. When $Y_{it} = \ln(MilExp_{it})$, this vector includes the natural logarithm of real GDP ($\ln(GDP_{it})$), the Polity V score ($Polity_{it}$) and its squared term ($Polity^2_{it}$), and the lagged dependent variable ($Y_{i,t-1}$). When $Y_{it} = MilShare_{it}$, the GDP control is excluded by construction to avoid collinearity and interpretational issues.

- $\alpha_i$ represents country fixed effects, which absorb all time-invariant, unobserved characteristics of state $i$.

- $\tau_t$ represents year fixed effects, which control for global shocks and trends common to all countries in year $t$[1].

- $\varepsilon_{it}$ is the idiosyncratic error term.

The inclusion of the interaction term alters the interpretation of our main coefficients. Specifically, $\beta_1$ captures the marginal effect of sanctions on military expenditure *strictly in peacetime* (when $War_{it} = 0$). The coefficient $\beta_2$ represents the effect of entering a war when *no sanctions* are imposed ($Sanctions_{it} = 0$). The coefficient on the interaction term, $\beta_3$, is our parameter of interest for heterogeneous effects: it represents the *additional* effect of sanctions that materializes only during armed conflicts. The total marginal effect of sanctions during wartime is given by the linear combination $(\beta_1 + \beta_3)$.

The coefficient of primary interest is $\beta_1$, which captures the marginal effect of sanctions on military expenditure in peacetime (when $War_{it} = 0$). The linear combination $(\beta_1 + \beta_2)$ represents the total effect of sanctions during wartime (when $War_{it} = 1$).

Throughout all baseline estimations, robust standard errors (HC1) are used.

While these TWFE models provide a strong baseline by controlling for a wide array of observable and unobservable (time-invariant) factors, we acknowledge that they do not fully resolve the fundamental issue of endogeneity. Specifically, the imposition of

[1]While the simultaneous inclusion of country fixed effects and a lagged dependent variable introduces Nickell (1981) bias, this bias is of order $O(1/T)$. Given the large time dimension of our panel ($T \approx 74$), this bias is considered as asymptotically negligible, precluding the need for GMM estimators (e.g., Arellano-Bond).

sanctions is rarely a random event; it may be driven by unobserved, time-varying factors (e.g., a sudden shift in a target's aggressive foreign policy) that simultaneously dictate its military expenditure, leading to omitted variable bias and/or reverse causality concerns. Consequently, while these OLS results are presented for comparative purposes, they should be interpreted cautiously as conditional correlations rather than definitive causal effects. To address this endogeneity, we will subsequently turn to our 2SLS strategy.

## 4.1 Baseline Results

Having established the baseline correlation between aggregate sanction pressure and military expenditure, we now disaggregate our main independent variable to examine whether the effects vary by the specific type of sanction imposed. Different forms of economic and political coercion — ranging from targeted arms embargoes to broad financial restrictions — impose different costs and signal varying degrees of resolve, which may elicit distinct responses from the target's defense establishment.

Table 3: Baseline OLS Estimations: Impact of Sanctions on Military Expenditure

| | **Dependent Variable:** $\ln(MilExp_{it})$ | | | | | | |
|---|---|---|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** | **(5)** | **(6)** | **(7)** |
| **Variables** | Pooled | Year FE | Country FE | TWFE | Full | TWFE Int. | Full Int. |
| Sanctions (Log Count) | 0.025*** | 0.082*** | 0.104*** | 0.099*** | 0.013*** | 0.099*** | 0.014*** |
| | (0.007) | (0.007) | (0.010) | (0.008) | (0.005) | (0.008) | (0.004) |
| War | 1.032*** | 0.870*** | 0.100** | 0.157*** | 0.030 | 0.166*** | 0.058** |
| | (0.064) | (0.069) | (0.048) | (0.047) | (0.031) | (0.057) | (0.025) |
| Sanctions × War | | | | | | -0.006 | -0.021 |
| | | | | | | (0.025) | (0.028) |
| $\ln(GDP_{it})$ | 0.990*** | 1.006*** | 0.637*** | 0.572*** | 0.067*** | 0.572*** | 0.067*** |
| | (0.009) | (0.010) | (0.019) | (0.035) | (0.013) | (0.035) | (0.013) |
| Lagged $\ln(MilExp_{it})$ | | | | | 0.884*** | | 0.884*** |
| | | | | | (0.012) | | (0.012) |
| Polity | | | | | 0.001 | | 0.001 |
| | | | | | (0.001) | | (0.001) |
| Polity$^2$ | | | | | -0.000 | | -0.000 |
| | | | | | (0.000) | | (0.000) |
| **Fixed Effects** | | | | | | | |
| Country FE | No | No | Yes | Yes | Yes | Yes | Yes |
| Year FE | No | Yes | No | Yes | Yes | Yes | Yes |
| **Summary Statistics** | | | | | | | |
| Observations | 7,855 | 7,855 | 7,855 | 7,855 | 4,918 | 7,855 | 4,918 |
| $R^2$ | 0.860 | 0.868 | 0.640 | 0.419 | 0.911 | 0.419 | 0.911 |

*Notes:* Robust standard errors are in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. The dependent variable is the natural logarithm of real military expenditure in constant 2023 US dollars. "Sanctions (Log Count)" is ln(1+Number of sanctioning states). The drop in observations in models (5) and (7) is due to the inclusion of the lagged dependent variable and Polity scores.

The initial OLS regressions indicate a robust positive correlation between the number of sanctioning states and military expenditure. This positive relationship persists even when controlling for domestic institutions (the Polity index), bureaucratic inertia (lagged dependent variable), and involvement in armed conflicts, including the interaction between war and sanctions. Notably, we find no statistically significant difference in this correlation between peacetime and wartime, which tentatively suggests a lack of heterogeneity in the sanctions' effect based on active military engagement. The persistence of these positive correlations underscores our initial concerns regarding omitted variable bias and necessitates the use of a properly instrumented, exogenous independent variable.

To gain a more nuanced understanding, we proceed to disaggregate the sanction measures by type.

Table 4 presents the results of estimating the most comprehensive baseline model (Specification 7, Table 3): TWFE, the full vector of controls, and the interaction term are included, separately for six categories of sanctions: Arms, Military, Trade, Financial, Travel, and Other. The sanction measure used here is, again, the log-transformed count of sanctioning states.

Table 4: Baseline OLS Estimations by Sanction Type (Specification 7), Table 3

| | **Dependent Variable:** $\ln(MilExp_{it})$ | | | | | |
|---|---|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** | **(5)** | **(6)** |
| **Variables** | Arms | Military | Trade | Financial | Travel | Other |
| Sanctions (Log Count) | 0.021*** | 0.022*** | 0.003 | 0.019*** | 0.022*** | 0.013* |
| | (0.005) | (0.006) | (0.006) | (0.005) | (0.007) | (0.007) |
| Sanctions × War | -0.028 | -0.017 | -0.014 | -0.013 | -0.033 | -0.009 |
| | (0.031) | (0.042) | (0.056) | (0.067) | (0.104) | (0.088) |
| War | 0.058** | 0.040 | 0.044* | 0.038 | 0.042** | 0.035 |
| | (0.022) | (0.024) | (0.024) | (0.030) | (0.021) | (0.024) |
| $\ln(GDP_{it})$ | 0.068*** | 0.071*** | 0.064*** | 0.069*** | 0.070*** | 0.065*** |
| | (0.012) | (0.013) | (0.013) | (0.013) | (0.013) | (0.013) |
| Lagged $\ln(MilExp_{it})$ | 0.882*** | 0.884*** | 0.889*** | 0.884*** | 0.886*** | 0.888*** |
| | (0.012) | (0.012) | (0.012) | (0.012) | (0.012) | (0.012) |
| Polity | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 | 0.001 |
| | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| Polity$^2$ | -0.000 | -0.000* | -0.000** | -0.000 | -0.000* | -0.000** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Country, Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 |
| $R^2$ | 0.911 | 0.911 | 0.910 | 0.911 | 0.911 | 0.911 |

*Notes:* Robust standard errors in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. The dependent variable in all models is the natural logarithm of real military expenditure. The primary independent variable "Sanctions (Log Count)" changes across columns to match the specific sanction type indicated in the column header. All models represent the fully specified baseline (TWFE + Controls + Interaction).

When breaking down the analysis, we attempt to capture correlations with the specific types of sanctions considered in this study. It can be observed that while most sanctions positively correlate with defense spending, trade and "other" sanctions yield statistically insignificant estimates close to zero. This might suggest that even the presence of a positive endogeneity bias is insufficient to drive the true effect of such sanctions into positive territory. Nevertheless, these variables are retained for further analysis to test alternative explanations. For instance, if certain sanctions (particularly the less radical ones) actually appease the target regime — prompting it to signal a lack of confrontational intent by reducing defense budgets — the expected bias could theoretically be negative rather than positive. In such a scenario, the OLS estimates would be biased downwards, masking the true dynamics.

Table 5: Baseline OLS Estimations by Sanction Type (GDP-Weighted Measure)

| | **Dependent Variable:** $\ln(MilExp_{it})$ | | | | | | |
|---|---|---|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** | **(5)** | **(6)** | **(7)** |
| **Variables** | All | Arms | Military | Trade | Financial | Travel | Other |
| Sanctions (Log GDP-Wgt) | 0.001* | 0.002*** | 0.001*** | -0.000 | 0.001*** | 0.002** | 0.001* |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.001) | (0.001) |
| Sanctions × War | -0.003 | -0.002 | -0.001 | -0.002 | -0.001 | -0.005 | -0.001 |
| | (0.002) | (0.003) | (0.003) | (0.004) | (0.003) | (0.012) | (0.006) |
| War | 0.073*** | 0.054** | 0.041* | 0.056*** | 0.042 | 0.045** | 0.035 |
| | (0.027) | (0.023) | (0.025) | (0.020) | (0.026) | (0.020) | (0.022) |
| $\ln(GDP_{it})$ | 0.065*** | 0.067*** | 0.066*** | 0.064*** | 0.066*** | 0.067*** | 0.065*** |
| | (0.013) | (0.012) | (0.013) | (0.013) | (0.013) | (0.013) | (0.013) |
| Lagged $\ln(MilExp_{it})$ | 0.889*** | 0.884*** | 0.888*** | 0.890*** | 0.886*** | 0.887*** | 0.888*** |
| | (0.012) | (0.012) | (0.012) | (0.012) | (0.012) | (0.012) | (0.012) |
| Polity | 0.000 | 0.001 | 0.000 | 0.000 | 0.001 | 0.001 | 0.001 |
| | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) |
| Polity$^2$ | -0.000* | -0.000 | -0.000** | -0.000** | -0.000 | -0.000* | -0.000** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| **Fixed Effects** | | | | | | | |
| Country FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Summary Statistics** | | | | | | | |
| Observations | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 |
| $R^2$ | 0.911 | 0.911 | 0.911 | 0.910 | 0.911 | 0.911 | 0.911 |

*Notes:* Robust standard errors in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. The dependent variable is the natural logarithm of real military expenditure. The primary independent variable "Sanctions (Log GDP-Wgt)" is constructed as $\ln(1 + \sum GDP_{sender,t})$, capturing the aggregate economic size of the sanctioning coalition. Column (1) aggregates all sanction types. Columns (2)-(7) disaggregate by specific sanction types. All models use the fully specified baseline formulation (TWFE + Controls + Interaction).

For a robustness check, we alternatively define sanction pressure based on the economic weight of the sender states. Formally, this GDP-weighted measure is calculated as the natural logarithm of one plus the sum of the real GDP of all countries imposing sanctions on target $i$ in year $t$: $\ln(1 + \sum GDP_{sender,t})$. This serves as a more direct proxy for the vulnerability of the target economy, capturing the extent to which major global economic powers are involved in the sanctioning coalition.

The results largely mirror those presented in the previous table. All sanction measures — except for trade and "other" sanctions (again, the latter likely due to sample size limitations) — remain significantly and positively correlated with military expenditure.

Table 6: Baseline OLS Estimations: Impact of Sanctions on Military Burden (Share of GDP)

| | **Dependent Variable: Military Expenditure / GDP** | | | | | | |
|---|---|---|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** | **(5)** | **(6)** | **(7)** |
| **Variables** | All | Arms | Military | Trade | Financial | Travel | Other |
| Sanctions (Log Count) | 0.002** | 0.002* | 0.003* | 0.003 | 0.002* | 0.001* | 0.001 |
| | (0.001) | (0.001) | (0.002) | (0.002) | (0.001) | (0.000) | (0.001) |
| Sanctions × War | -0.001 | -0.002 | -0.003 | -0.001 | -0.004 | -0.003 | -0.003 |
| | (0.002) | (0.002) | (0.003) | (0.004) | (0.004) | (0.006) | (0.005) |
| War | 0.012*** | 0.012*** | 0.013*** | 0.012*** | 0.013*** | 0.012*** | 0.012*** |
| | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| Lagged $MilShare_{it}$ | 0.658*** | 0.658*** | 0.656*** | 0.661*** | 0.659*** | 0.665*** | 0.664*** |
| | (0.210) | (0.209) | (0.207) | (0.208) | (0.208) | (0.211) | (0.212) |
| Polity | -0.000* | -0.000** | -0.000** | -0.000 | -0.000* | -0.000** | -0.000** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| Polity$^2$ | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) | (0.000) |
| **Fixed Effects** | | | | | | | |
| Country FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| **Summary Statistics** | | | | | | | |
| Observations | 5,086 | 5,086 | 5,086 | 5,086 | 5,086 | 5,086 | 5,086 |
| $R^2$ | 0.485 | 0.486 | 0.488 | 0.486 | 0.486 | 0.482 | 0.482 |

*Notes:* Robust standard errors in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. The dependent variable is military expenditure as a share of GDP ($MilShare_{it}$). By construction, the real GDP control variable is excluded from these specifications to avoid endogeneity and interpretational issues. The primary independent variable "Sanctions (Log Count)" is ln(1 + Number of sanctioning states). All models use the fully specified baseline formulation (TWFE + Controls + Interaction).

Finally, altering the dependent variable to the military burden (the share of military expenditure in the target's GDP) yields consistent OLS results. Similar to the models using absolute spending, sanctions (aggregate, arms, military, financial, and travel) exhibit a positive and significant correlation (albeit sometimes only at the 10% level) with an increase in the military burden during peacetime. Taken at face value, this implies that target states do not merely increase absolute defense spending but actively reallocate budget priorities toward the military sector.

To summarize, the baseline OLS regressions suggest that international sanctions — with the notable exception of trade restrictions — are associated with an increase in military expenditure, both in absolute terms (increased armament) and as a share of the national economy. Furthermore, this correlational effect does not appear to depend on whether the target state is actively engaged in a war.

However, as previously noted, these OLS estimates suffer from fundamental methodological flaws — primarily reverse causality and omitted variable bias arising from the non-random nature of sanction imposition. Consequently, the observed positive correlations cannot be interpreted as causal relationships. To isolate the true causal impact of sanctions on military expenditure and address the biases discussed above, we must turn to an IV strategy.

# 5 Empirical Strategy: Instrumental Variables Approach

While the baseline OLS estimates presented in Section 4 offer valuable insights into the conditional correlations between international sanctions and military expenditure, they cannot be interpreted causally. The fundamental econometric issue in this context is the endogeneity of sanction imposition.

Sanctions are not randomly assigned exogenous shocks. Rather, they are strategic policy tools deployed in response to the specific behavior and characteristics of the target state. A country that is actively militarizing, pursuing an aggressive foreign policy, or violating international norms is simultaneously more likely to increase its defense budget and more likely to attract international sanctions. Consequently, the error term in our baseline Equation (1) is likely correlated with the sanction variable ($Cov(Sanctions_{it}, \varepsilon_{it}) \neq 0$), leading to omitted variable bias. Furthermore, the prospect of facing sanctions might induce a target state to preemptively adjust its military spending, introducing reverse causality. In such scenarios, standard OLS estimates are likely biased upwards, capturing the underlying militarization trend rather than the true, isolated effect of the economic coercion.

To address these endogeneity concerns and isolate the causal impact of sanctions, we employ an IV strategy, specifically a 2SLS approach. Following the methodological framework outlined in an article examining the impact of sanctions on GDP (Kwon et al., 2022), our instrument is constructed using a "zero-stage" dyadic model to predict the probability of sanction imposition based on strictly exogenous factors.

## 5.1 Predicting Sanctions

In the zero-stage, we construct a dyadic dataset where the unit of observation is a sender-target-year ($i$-$j$-$t$) triad. We model the probability that sender $i$ imposes a sanction on target $j$ in year $t$ using a linear probability model (instead of logistic regression) to avoid the "perfect separation" problem common in non-linear models with rare events and high-dimensional fixed effects. Although probabilities are traditionally modeled using Probit/Logit (and it was proposed in the original article (Kwon et al., 2022), where the problem of the rarity of sanctions of certain types was not so acute), a Linear Probability Model (LPM) is used due to the problem of perfect separation. Since specific sanctions (e.g., financial ones) are rare events, including fixed effects in nonlinear models leads to a loss of convergence. LPM provides consistent estimates in the first step of 2SLS without these problems (Angrist and Pischke, 2008). The zero-stage equation is specified as follows[2]:

[2]Equation (2) is estimated using OLS with robust standard errors, yielding predicted probabilities $\widehat{Sanction}_{ijt}$ that may technically fall outside the [0,1] interval, which is unproblematic for the first stage of a 2SLS procedure.

$$Sanction_{ijt} = \delta_0 + \delta_1 Aggr_{i,-j,t-1} + \delta_2 VotingDist_{ij,t-1} + \delta_3 (Aggr \times VoteDist) \theta' \mathbf{G}_{ij} + \tau_t + \lambda_j + \nu_{ijt} \quad (2)$$

Where:

- $Sanction_{ijt}$ is a binary indicator equalling 1 if sender $i$ sanctions target $j$ in year $t$, and 0 otherwise.
- $Aggr_{i,-j,t-1}$ is our core exogenous instrument. It measures the general propensity of the sender state to utilize sanctions. To ensure exogeneity with respect to target $j$, a "leave-one-out" approach is employed: it is calculated as the log-transformed number of sanctions sender $i$ imposed on *all other countries except* $j$. We use the one-year lag to avoid simultaneity bias.
- $VotingDist_{ij,t-1}$ captures the lagged political distance between the dyad, measured by the average absolute difference in their UN General Assembly voting records. (UN Dag Hammarskjöld Library, 2024). The political distance between the sender $i$ and the target $j$ is defined as the mean absolute difference in their voting choices across all shared UN General Assembly resolutions in a given year:
$$VotingDist_{ijt} = \frac{1}{R_{ijt}} \sum_{r=1}^{R_{ijt}} |V_{irt} - V_{jrt}| \quad (3)$$
where $V_{irt}$ and $V_{jrt}$ denote the voting choices of countries $i$ and $j$ on resolution $r$, coded as 1 (Yes), 0 (No), or 0.5 (Abstain or Absent), and $R_{ijt}$ is the total number of resolutions during year $t$.
- We include an interaction term between lagged aggressiveness and lagged voting distance to capture potential non-linearities in sanctioning behavior.
- $\mathbf{G}_{ij}$ is a vector of strictly exogenous, time-invariant gravity controls from the CEPII database (Mayer and Zignago, 2011), including log geographic distance, contiguity, common official language, and colonial links.
- $\tau_t$ and $\lambda_j$ represent year and target-country fixed effects, respectively.

The detailed results of the zero-stage estimations for representative sanction types are provided in Appendix Table 16, confirming high significance of these instruments as predictors of the probability of sanction imposition. Equation 2 is estimated iteratively (separately) for each sanction type (arms, military, trade, etc.) to obtain type-specific predicted probabilities. Thus, we obtain the predicted probability of a sanction occurring, $\widehat{Sanction}_{ijt}$.

## 5.2 2SLS Estimation

Having obtained the exogenous predicted probabilities at the dyadic (i.e. sender-target-year level), we aggregate them to the target-year level to construct our instrument ($Z_{jt}$). Depending on the sanction measure employed (count or GDP-weighted), the instrument is calculated as either the log-transformed sum of the predicted probabilities ($\ln(1 + \sum_i \widehat{Sanction}_{ijt})$) or the log-transformed sum of the sender's GDP weighted by these probabilities ($\ln(1 + \sum_i [\widehat{Sanction}_{ijt} \times GDP_{it}])$).

With our valid instrument $Z_{jt}$ calculated, we proceed to the standard 2SLS estimation. To test for heterogeneous effects during wartime, we instrument both the sanction measure and its interaction with the war dummy.

The **First Stage** equations are defined as[3]:

$$Sanctions_{jt} = \pi_{10} + \pi_{11} Z_{jt} + \pi_{12}(Z_{jt} \times War_{jt}) + \phi_1' \mathbf{X}_{jt} + \alpha_j + \tau_t + u_{1jt} \tag{4}$$

$$(Sanctions_{jt} \times War_{jt}) = \pi_{20} + \pi_{21} Z_{jt} + \pi_{22}(Z_{jt} \times War_{jt}) + \phi_2' \mathbf{X}_{jt} + \alpha_j + \tau_t + u_{2jt} \tag{5}$$

where $Z_{jt}$ and $(Z_{jt} \times War_{jt})$ serve as the excluded instruments. The F-statistics from these first-stage regressions are monitored to ensure the instruments are sufficiently strong (exceeding the threshold of 10) and relevant.

Finally, the **Second Stage** equation estimates the causal impact by substituting the endogenous variables with their predicted values ($\widehat{Sanctions}_{jt}$ and $\widehat{Sanctions \times} War_{jt}$) obtained from the first stage:

$$Y_{jt} = \gamma_0 + \gamma_1 \widehat{Sanctions}_{jt} + \gamma_2 \widehat{Sanctions \times} War_{jt} + \gamma_3 War_{jt} + \rho' \mathbf{X}_{jt} + \alpha_j + \tau_t + \eta_{jt} \tag{6}$$

By relying on the exogenous variation in sanction pressure isolated by our instruments, Equation (6) allows us to interpret $\gamma_1$ and the linear combination $(\gamma_1 + \gamma_2)$ as the unbiased, causal effects of international sanctions on military expenditure both in peacetime and wartime.

## 5.3 Gravity-Based Weights

As will be shown below, the study of trade-related measures is highly sensitive to the choice of weights. However, a sender's share of global GDP is a suboptimal proxy for measuring the sensitivity of a target to specific restrictions. Factors such as geographical location, internal population distribution, and the development of transport infrastructure mean that every country possesses a unique external trade structure that often diverges from the simple GDP rankings of its partners. While weighting sanctions

[3]Following (Wooldridge, 2010), the interaction between sanctions and war is treated as a separate endogenous regressor. We avoid the so-called 'forbidden regression' — which would involve simply multiplying the first-stage prediction of sanctions by the war dummy — to ensure that our second-stage estimates remain consistent and the standard errors are correctly estimated.

by actual bilateral trade volumes would provide a more direct measure of economic vulnerability, such volumes are fundamentally endogenous. Trade flows are directly influenced by the sanctions themselves, as well as by the underlying state of international relations, which simultaneously dictates both the probability of sanction imposition and the observed intensity of commercial exchange.

To circumvent this fundamental endogeneity problem, we adopt a structural gravity-based IV approach, a methodology widely utilized in the empirical trade and macroeconomic literature to construct exogenous measures of trade exposure (Frankel and Romer, 1999; Santos Silva and Tenreyro, 2006; Feyrer, 2019). The core idea is to estimate a gravity model of trade to generate predicted bilateral trade volumes that are driven strictly by exogenous geographic and historical determinants, thereby purging the measure of political shocks and actual sanction policies.

In a "Stage -1" procedure, bilateral trade flows are estimated using the Poisson Pseudo-Maximum Likelihood (hereinafter PPML) estimator, which robustly handles zero trade flows and heteroskedasticity(Santos Silva and Tenreyro, 2006). The regression coefficients for these gravity models, which show a high (at first glance) degree of fit which, however, is in line with the theoretical expectations of the gravity framework, are reported in Appendix Table 15. We estimate separate gravity equations for exports and imports to capture the directional dependence of the target country. The gravity equations take the following form:

$$TradeFlow_{ji,t} = \exp\left[\theta_1 \ln(GDP_{it}) + \theta_2 \ln(GDP_{jt}) + \theta_3 \ln(Dist_{ij}) + \lambda' \mathbf{G}_{ij}\right] \upsilon_{ji,t} \tag{7}$$

$$TradeFlow_{ij,t} = \exp\left[\psi_1 \ln(GDP_{it}) + \psi_2 \ln(GDP_{jt}) + \psi_3 \ln(Dist_{ij}) + \delta' \mathbf{G}_{ij}\right] \epsilon_{ij,t} \tag{8}$$

Where $TradeFlow_{ji,t}$ represents the exports from target $j$ to sender $i$, and $TradeFlow_{ij,t}$ represents imports to target $j$ from sender $i$. The predictors include the economic mass of both countries ($\ln(GDP)$) and a vector of strictly exogenous, time-invariant geographic and historical bilateral frictions ($\mathbf{G}_{ij}$), including contiguous borders, common official language, and colonial links.

From equations (7) and (8), we extract the predicted trade volumes, $\widehat{TradeFlow}_{ji,t}$ and $\widehat{TradeFlow}_{ij,t}$. These fitted values represent the "natural" level of trade between the dyad, dictated solely by geography and size, completely independent of current political relations or sanctions.

Finally, we calculate the exogenous, gravity-predicted trade weights for each sender $i$ relative to target $j$:

$$\widehat{Weight}_{ijt}^{Export} = \frac{\widehat{TradeFlow}_{ji,t}}{\sum_i \widehat{TradeFlow}_{ji,t}} \tag{9}$$

$$\widehat{Weight}_{ijt}^{Import} = \frac{\widehat{TradeFlow}_{ij,t}}{\sum_i \widehat{TradeFlow}_{ij,t}} \tag{10}$$

These exogenous shares ($\widehat{Weight}_{ijt}$) are then multiplied by the actual sanction indicators on the Second Stage, and by the zero-stage LPM-predicted probabilities to form our instruments on the First Stage. This ensures that our measurement of sanction intensity strictly reflects the target's exogenous structural vulnerability to the sanctioning coalition.

## 5.4 Results

Table 7 presents the second-stage results for four primary specifications: absolute military expenditure (log-transformed) and military expenditure as a share of GDP, utilizing both the log-count and the GDP-weighted measures of sanction pressure. For ease of interpretation, the coefficients are reported such that the total effect of sanctions during wartime is directly observable, eliminating the need to manually sum the main effect and the interaction term.[4] We also report first-stage F-statistics, where values exceeding 10 indicate sufficient instrument relevance, and $p$-values from F-tests evaluating the equality of peacetime and wartime coefficients to detect potential heterogeneity in the causal effect.

First, the diagnostic indicators for instrument strength significantly exceed the conventional Stock-Yogo threshold ($F > 10$)(Stock and Yogo, 2005). This allows us to reject the weak instrument hypothesis, suggesting that our zero-stage LPM effectively isolates exogenous variation in sanction intensity. The zero-stage analysis confirms the validity of our exogenous predictors: as expected, lagged sender aggressiveness and political distance (UNGA voting distance) positively and significantly predict the likelihood of sanctions in all four specifications.

However, despite the strength of the instruments, none of the sanction coefficients — pertaining to either peacetime or wartime — reach statistical significance. This leads to the conclusion that the positive correlations identified in the baseline OLS models were driven entirely by endogeneity bias. Target states do not appear to increase their military budgets in response to exogenous sanction shocks; rather, sanctions tend to be imposed on countries already exhibiting aggressive behavior or undergoing militarization for unobserved internal reasons. This null effect is robust across both the prioritization of defense resources (share of GDP) and the absolute volume of defense spending. Furthermore, F-tests for coefficient equality fail to reject the null hypothesis, indicating that the causal impact of aggregate

[4] Specifically, we estimate the model using $(1 - War) \times Sanctions$ and $War \times Sanctions$ to obtain direct point estimates and standard errors for both regimes.

sanctions remains statistically indistinguishable from zero regardless of whether the target country is engaged in an active conflict.

Table 7: 2SLS Estimations: The Causal Impact of Aggregate Sanctions

| | **DV:** $\ln(MilExp_{it})$ **(Absolute)** | | **DV:** $MilShare_{it}$ **(Share of GDP)** | |
|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** |
| **Variables** | Log Count | Log GDP-Wgt | Log Count | Log GDP-Wgt |
| Sanctions (Peacetime) | 0.040 | 0.005 | 0.002 | 0.000 |
| | (0.038) | (0.005) | (0.003) | (0.000) |
| Sanctions (Wartime) | -0.194 | -0.027 | -0.005 | -0.001 |
| | (0.166) | (0.022) | (0.017) | (0.003) |
| War | 0.108 | 0.148 | 0.011* | 0.013 |
| | (0.083) | (0.115) | (0.006) | (0.011) |
| $\ln(GDP_{it})$ | 0.072*** | 0.069*** | | |
| | (0.015) | (0.014) | | |
| Polity | 0.001 | 0.001 | -0.000* | -0.000 |
| | (0.001) | (0.001) | (0.000) | (0.000) |
| Polity$^2$ | -0.000 | -0.000 | 0.000 | 0.000* |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| Lagged Dependent Var. | 0.873*** | 0.876*** | 0.654*** | 0.660*** |
| | (0.022) | (0.020) | (0.231) | (0.227) |
| **Fixed Effects** | | | | |
| Country FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| **IV Diagnostics & Stats** | | | | |
| First-Stage F-Statistic | **22.23** | **46.61** | **23.14** | **37.51** |
| Equality Test ($p$-value)† | 0.205 | 0.192 | 0.669 | 0.700 |
| Observations | 4,918 | 4,918 | 4,965 | 4,965 |
| $R^2$ (uncentered) | 0.919 | 0.919 | 0.484 | 0.484 |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. Models are estimated using 2SLS on demeaned data. All models include the lagged dependent variable ($\ln(MilExp_{i,t-1})$ in columns 1-2, and $MilShare_{i,t-1}$ in columns 3-4) as a control. The instrument is derived from a zero-stage LPM utilizing lagged leave-one-out sender aggressiveness and its interaction with UNGA voting distance.

† The Equality Test reports the $p$-value for the $F$-test of the null hypothesis that the effect of sanctions in peacetime equals the effect in wartime ($H_0 : \gamma_{peace} = \gamma_{war}$).

Now it is important to pay attention to the potential heterogeneity of these effects by disaggregating sanction pressure into specific categories. Following the classification established in the Global Sanctions Data Base (GSDB) (Felbermayr et al., 2020), we distinguish between six primary types of restrictive measures:

- **Arms Sanctions**: Prohibitions on the supply, sale, or transfer of weapons, ammunition, military vehicles, and related equipment.
- **Military Assistance Sanctions**: Restrictions on providing monetary aid, technical training, or logistical support intended for the target's armed forces.
- **Trade Sanctions**: Measures aiming to restrain commercial interactions through partial or complete bans on exports and/or imports of goods and services.
- **Financial Sanctions**: Restrictions on financial transactions, including asset freezes, investment prohibitions, and limits on access to international credit or payment systems.
- **Travel Sanctions**: Limitations on the geographical movement of specific individuals (e.g., travel bans for elites) or broader groups between the sender and target countries.
- **Other Sanctions**: A residual category encompassing diplomatic measures, such as the severance of relations, as well as specific logistical constraints like flight or harbor restrictions.

The 2SLS estimation results for each specific type are presented in Table 8 (utilizing the log-count measure) and Table 9 (utilizing the GDP-weighted measure).

It is important to note that a robust instrument was not obtained for all sanction types: instruments for tourism and financial sanctions proved weak. This is likely due to the fact that such smart sanctions are targeted policy instruments that are weakly predicted by structural gravity and macropolitical (UNGA) factors compared to broad trade embargoes. Specifications where the first-stage F-statistic falls below the conventional threshold of 10 are indicated in italics. Because these estimates are likely compromised by weak instrument bias, we refrain from a causal interpretation of these specific coefficients and instead focus on the categories where the instruments demonstrate sufficient relevance.

Our findings indicate that neither arms embargoes nor sanctions categorized as "Other" exert a significant causal impact on military expenditure levels. In the case of arms sanctions, the lack of a budgetary response likely reflects the target state's ability to secure alternative suppliers or pursue domestic import substitution. While such constraints may degrade actual military capabilities by forcing the adoption of sub-optimal technologies, they do not appear to translate into significant shifts in aggregate defense spending. Regarding "Other" sanctions, their impact is likely too marginal to serve as an effective coercive signal; even in the baseline OLS specifications, these measures showed only borderline significance, suggesting they lack the necessary leverage to induce substantial changes in a recipient state's military resource allocation.

Table 8: 2SLS Estimations by Sanction Type (Measure: Log Count of Senders)

| | (1) Arms | (2) Military | (3) Trade | (4) Financial | (5) Travel | (6) Other |
|---|---|---|---|---|---|---|
| **Panel A: Dependent Variable - $\ln(MilExp_{it})$ (Absolute Expenditure)** | | | | | | |
| Sanctions (Peacetime) | 0.028 | -0.072 | 0.063 | -0.032 | -0.061 | 0.083 |
| | (0.030) | (0.133) | (0.041) | (0.141) | (0.152) | (0.058) |
| Sanctions (Wartime) | -0.166 | -1.443** | -0.149 | -0.997** | -1.665*** | -0.218 |
| | (0.188) | (0.571) | (0.218) | (0.423) | (0.628) | (0.178) |
| *IV Diagnostics:* | | | | | | |
| First-Stage F-Statistic | **33.95** | **61.12** | *8.23* | *4.22* | *0.01* | **12.17** |
| Equality Test ($p$-value)† | 0.338 | 0.012 | 0.329 | 0.017 | 0.007 | 0.113 |
| Observations | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 |
| **Panel B: Dependent Variable - $MilShare_{it}$ (Share of GDP)** | | | | | | |
| Sanctions (Peacetime) | 0.002 | -0.036 | 0.002 | -0.041* | -0.097** | 0.001 |
| | (0.002) | (0.025) | (0.004) | (0.022) | (0.043) | (0.006) |
| Sanctions (Wartime) | -0.000 | -0.086 | -0.004 | -0.092*** | -0.184*** | -0.003 |
| | (0.017) | (0.055) | (0.032) | (0.030) | (0.051) | (0.026) |
| *IV Diagnostics:* | | | | | | |
| First-Stage F-Statistic | **36.74** | **59.60** | *7.84* | *3.10* | *0.01* | **12.05** |
| Equality Test ($p$-value)† | 0.917 | 0.422 | 0.851 | 0.181 | 0.087 | 0.886 |
| Observations | 4,965 | 4,965 | 4,965 | 4,965 | 4,965 | 4,965 |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. All models are estimated using 2SLS on demeaned data and include Country and Year Fixed Effects. The primary independent variable is $\ln(1+\text{Count of sanctioning states})$. Panel A includes controls for $\ln(GDP)$, War, Polity, Polity$^2$, and lagged $\ln(MilExp)$. Panel B excludes $\ln(GDP)$ to avoid collinearity and includes lagged $MilShare$. Models with a First-Stage F-statistic below 10 are reported in italics to highlight the potential presence of weak instruments.

† The Equality Test reports the $p$-value for the $F$-test of the null hypothesis that the effect of sanctions in peacetime equals the effect in wartime ($H_0 : \gamma_{peace} = \gamma_{war}$).

Table 9: 2SLS Estimations by Sanction Type (Measure: GDP-Weighted)

| | **(1)** **Arms** | **(2)** **Military** | **(3)** **Trade** | **(4)** **Financial** | **(5)** **Travel** | **(6)** **Other** |
|---|---|---|---|---|---|---|
| **Panel A: Dependent Variable - $\ln(MilExp_{it})$ (Absolute Expenditure)** | | | | | | |
| Sanctions (Log GDP-Wgt) | 0.004 | -0.003 | 0.006 | *0.008* | *-0.011* | 0.006 |
| | (0.004) | (0.009) | (0.006) | *(0.027)* | *(0.024)* | (0.004) |
| Sanctions (Wartime) | -0.023 | -0.171** | -0.044 | *-0.053* | *-0.210**** | -0.013 |
| | (0.023) | (0.075) | (0.030) | *(0.045)* | *(0.082)* | (0.013) |
| *IV Diagnostics:* | | | | | | |
| First-Stage F-Statistic | **55.69** | **31.40** | **18.68** | *0.11* | *0.04* | **53.22** |
| Equality Test ($p$-value)† | 0.265 | 0.025 | 0.063 | 0.060 | 0.009 | 0.174 |
| Observations | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 | 4,918 |
| **Panel B: Dependent Variable - $MilShare_{it}$ (Share of GDP)** | | | | | | |
| Sanctions (Log GDP-Wgt) | 0.000 | -0.002 | -0.000 | *-0.003** | *-0.008*** | 0.000 |
| | (0.000) | (0.002) | (0.001) | *(0.002)* | *(0.004)* | (0.000) |
| Sanctions (Wartime) | -0.000 | -0.004 | -0.003 | *-0.007*** | *-0.021**** | 0.000 |
| | (0.002) | (0.007) | (0.004) | *(0.003)* | *(0.005)* | (0.002) |
| *IV Diagnostics:* | | | | | | |
| First-Stage F-Statistic | **57.06** | **32.45** | **17.54** | *0.19* | *0.02* | **54.93** |
| Equality Test ($p$-value)† | 0.796 | 0.767 | 0.542 | 0.211 | 0.016 | 0.974 |
| Observations | 4,965 | 4,965 | 4,965 | 4,965 | 4,965 | 4,965 |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. All models are estimated using 2SLS on demeaned data and include Country and Year Fixed Effects, as well as the full set of controls detailed in Table 6. The primary independent variable is $\ln(1 + \sum GDP_{sender,t})$, capturing the aggregate economic size of the sanctioning coalition. Models with a First-Stage F-statistic below 10 are reported in italics to highlight the presence of weak instruments.

† The Equality Test reports the $p$-value for the $F$-test of the null hypothesis that the effect of sanctions in peacetime equals the effect in wartime ($H_0 : \gamma_{peace} = \gamma_{war}$).

Military assistance sanctions warrant a distinct analysis due to their strong performance under a robust instrument. We identify a significant heterogeneous effect (equality test $p$-value = 0.012), wherein the impact of these measures is conditional on the target's involvement in armed conflict. In peacetime, these sanctions remain statistically indistinguishable from zero. However, for states engaged in active warfare, military sanctions induce a substantial causal reduction in absolute military expenditure.

This pattern suggests that targeted restrictions — such as the blockage of critical components or spare parts — impose physical constraints on procurement. Consequently, the target state is deprived of the ability to allocate resources effectively to its defense sector, even if a strong domestic impetus for militarization persists.

Notably, this causal contraction applies only to absolute spending levels and does not result in a corresponding reduction in the military burden (military expenditure as a share of GDP). A plausible explanation is that by preventing defense outlays, these sanctions inhibit the fiscal "heating up" characteristic of war economies. As a result, military sanctions—often deployed in tandem with broader restrictive measures—exert a decisive influence on the target's overall economic trajectory during wartime. Ultimately, the evidence demonstrates that military sanctions are effective in degrading the defense capabilities of warring states by creating hard resource bottlenecks.

Table 10: 2SLS Estimations: Trade Sanctions (Count and GDP-Weighted Measures)

| | **Panel A: DV = $\ln(MilExp_{it})$** | | | | **Panel B: DV = $MilShare_{it}$** | | | |
|---|---|---|---|---|---|---|---|---|
| | EXPORT | | IMPORT | | EXPORT | | IMPORT | |
| **Sanction Type** | Peace | War | Peace | War | Peace | War | Peace | War |
| **SUB-PANEL 1: Measure = Log Count of Senders** | | | | | | | | |
| **Any Trade** | -0.239 | -1.000 | 0.062* | -0.040 | -0.060** | -0.143*** | 0.001 | 0.012 |
| | (0.706) | (0.775) | (0.035) | (0.175) | (0.030) | (0.036) | (0.004) | (0.028) |
| *F-stat / Eq. Test (p)* | *0.54 / 0.083* | | *7.31 / 0.569* | | *0.81 / 0.028* | | *6.95 / 0.732* | |
| **Complete Only** | 1.136 | -2.162* | 0.054* | 0.022 | -0.157 | -0.269** | 0.002 | 0.020 |
| | (1.265) | (1.163) | (0.030) | (0.130) | (0.104) | (0.114) | (0.003) | (0.020) |
| *F-stat / Eq. Test (p)* | *0.81 / 0.074* | | *8.45 / 0.821* | | *1.03 / 0.535* | | *8.11 / 0.424* | |
| **SUB-PANEL 2: Measure = Log GDP-Weighted** | | | | | | | | |
| **Any Trade** | 0.028 | -0.040 | 0.006 | -0.027 | 0.001 | -0.003 | -0.000 | -0.001 |
| | (0.029) | (0.041) | (0.007) | (0.030) | (0.002) | (0.004) | (0.001) | (0.005) |
| *F-stat / Eq. Test (p)* | *2.69 / 0.089* | | **16.49 / 0.278** | | *2.56 / 0.316* | | **15.35 / 0.954** | |
| **Complete Only** | 0.059 | -0.033 | 0.010* | -0.010 | 0.004 | 0.002 | 0.001 | 0.001 |
| | (0.060) | (0.060) | (0.006) | (0.027) | (0.004) | (0.006) | (0.000) | (0.003) |
| *F-stat / Eq. Test (p)* | *1.57 / 0.286* | | **18.20 / 0.506** | | *1.41 / 0.797* | | **17.45 / 0.802** | |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. All models are estimated using 2SLS on demeaned data with Country and Year FEs. Sub-Panel 1 uses the count of sanctioning states; Sub-Panel 2 weights sanctions by the GDP of sending states. Panel A and B use different dependent variables as indicated. Controls include War, Polity, Polity$^2$, lagged DV, and $\ln(GDP)$ (in Panel A). F-stat refers to the first-stage F-statistic of the instrument. Eq. Test (p) is the p-value for the null hypothesis of equality between peacetime and wartime coefficients. Strong instruments (F > 10) are highlighted in bold.

However, trade sanctions aggregation requires careful examination. While a log-count measure of sanctioning states yields a weak instrument, the instrument becomes robust when aggregate sanction pressure is weighted by the GDP of the sender states. In these specifications, the causal effect of trade sanctions remains statistically insignificant during both peacetime and wartime. This indicates that aggregate trade measures are inherently heterogeneous, encompassing export and import restrictions of varying intensity and coverage. Furthermore, it highlights that trade-related pressure is highly sensitive to the economic weight of the senders, as the perceived cost to the target state is directly tied to the significance of the sanctioning economies.

To address the potential heterogeneity within this category, more granular data on the specific types of trade restrictions is utilized, provided by the GSDB (Felbermayr et al., 2020). We distinguish between the following four sub-types[5]:

- **Partial Export Sanctions**: Restrictions or selective bans on specific goods or particular sectors of trade exported from the sender to the target country.
- **Complete Export Sanctions**: Comprehensive prohibitions on the export of all goods and services from the sender to the target country.
- **Partial Import Sanctions**: Restrictions or selective bans on specific goods or particular sectors of trade imported from the target country to the sender.
- **Complete Import Sanctions**: Comprehensive prohibitions on the import of all goods and services from the target country to the sender.

Table 10 presents the results of models estimated for export and import sanctions separately. We further distinguish between specifications where sanctions are counted regardless of their stringency ("Any, i.e. partial or complete restrictions) and specifications that account only for "Complete" restrictions.

These disaggregated results emphasize the importance of differentiating sanctions by both their direction and their intensity. Specifications involving import sanctions, as well as all models that rely on a simple count of senders without GDP-weighting, fail to provide reliable instruments. Even in specifications where the instruments are robust, significant causal effects are largely absent. The only marginal exception is a borderline positive effect of complete import sanctions during peacetime; however, this result is not sufficiently robust to constitute conclusive evidence.

[5]NB: Hereinafter "Export / Import" refers to the direction of trade from the sender's perspective (i.e., Export sanctions restrict the target's imports).

Table 11: 2SLS Estimations: Combined Impact of Partial and Complete Trade Sanctions

| | Direction: EXPORT | | | | Direction: IMPORT | | | |
|---|---|---|---|---|---|---|---|---|
| | Log Count | | Log GDP-Wgt | | Log Count | | Log GDP-Wgt | |
| **Variables** | Peace | War | Peace | War | Peace | War | Peace | War |
| **Panel A: Dependent Variable - $\ln(MilExp_{it})$ (Absolute Expenditure)** | | | | | | | | |
| Partial Sanctions | 0.784 | 0.691 | -0.012 | -0.278*** | -0.241 | -1.988*** | -0.006 | -0.071 |
| | (0.746) | (0.970) | (0.052) | (0.101) | (0.409) | (0.704) | (0.012) | (0.043) |
| Complete Sanctions | 1.791 | -0.936 | 0.003 | 0.449* | 0.008 | -0.210 | 0.007 | -0.038 |
| | (1.678) | (0.674) | (0.052) | (0.255) | (0.051) | (0.131) | (0.006) | (0.028) |
| *F-stats (Part / Compl)* | *0.95 / 28.98* | | **30.81 / 27.45** | | *7.75 / 8.47* | | *6.65 / 18.18* | |
| **Panel B: Dependent Variable - $MilShare_{it}$ (Share of GDP)** | | | | | | | | |
| Partial Sanctions | -0.032 | -0.122*** | -0.009** | -0.018 | -0.094* | -0.291*** | -0.005* | -0.010** |
| | (0.038) | (0.041) | (0.004) | (0.015) | (0.055) | (0.087) | (0.003) | (0.005) |
| Complete Sanctions | -0.104 | -0.124** | 0.009** | 0.011 | -0.008 | -0.009 | -0.001 | -0.002 |
| | (0.161) | (0.060) | (0.004) | (0.035) | (0.006) | (0.020) | (0.001) | (0.004) |
| *F-stats (Part / Compl)* | *0.96 / 15.46* | | **29.84 / 28.69** | | *8.22 / 8.12* | | *6.92 / 17.46* | |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. All models estimated via 2SLS on demeaned data. Partial and Complete sanctions for the respective direction are included simultaneously. Controls include War, Polity, Polity$^2$, lagged DV, and $\ln(GDP)$ (in Panel A). F-stats represent the first-stage joint significance tests for the respective endogenous variables. Strong instruments (F > 10) for both types are in bold.

We subsequently estimate a combined specification that incorporates both partial and complete sanctions for a specific trade direction (export or import) within a single model. This approach yields four primary independent variables per specification (accounting for the wartime interaction) and is reported in Table 11. These results clarify the dynamics of export sanctions, particularly in GDP-weighted specifications. During wartime, the imposition of partial (sectoral) export sanctions leads to a significant reduction in military expenditure, while complete sanctions produce a nearly opposite result, reaching marginal levels of significance. In peacetime, partial sanctions appear to diminish a state's propensity for defense spending, shifting budgetary priorities and potentially achieving a degree of appeasement. In contrast, complete sanctions exert the opposite effect, increasing military spending in relative terms but not in absolute ones. This suggests that as the GDP (the overall "pie") contracts, the target regime protects the military's share of that budget, likely interpreting comprehensive restrictions as a high-intensity threat signal.

Table 12: 2SLS Estimations: Trade Sanctions Weighted by Predicted Gravity Flows

| | **Panel A: DV** = $\ln(MilExp_{it})$ | | | | **Panel B: DV** = $MilShare_{it}$ | | | |
|---|---|---|---|---|---|---|---|---|
| | EXPORT | | IMPORT | | EXPORT | | IMPORT | |
| **Sanction Type** | Peace | War | Peace | War | Peace | War | Peace | War |
| **Any Trade** | 0.447 | -4.557** | 0.107 | -3.747 | -0.198 | -0.797*** | -0.017 | -0.319 |
| | (1.217) | (2.150) | (0.185) | (2.850) | (0.140) | (0.190) | (0.027) | (0.518) |
| *F-stat / Eq. Test (p)* | *7.46 / 0.037* | | **36.15 / 0.175** | | *9.43 / 0.009* | | **34.71 / 0.567** | |
| **Complete Only** | 6.796 | -28.908 | 0.147 | -3.228 | -1.948** | -2.828* | 0.002 | -0.073 |
| | (18.441) | (25.582) | (0.149) | (5.221) | (0.990) | (1.510) | (0.011) | (0.726) |
| *F-stat / Eq. Test (p)* | *0.79 / 0.121* | | **43.97 / 0.519** | | *1.03 / 0.648* | | **43.29 / 0.918** | |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. All models are estimated using 2SLS on demeaned data and include Country and Year Fixed Effects. Sanction measures are weighted by exogenous bilateral trade shares predicted via a PPML gravity model. Following the mechanism of economic vulnerability, Export sanctions are weighted by the target's import dependence, and Import sanctions by the target's export dependence. Panel A includes full controls ($\ln(GDP)$, War, Polity, Polity$^2$, lagged $\ln(MilExp)$). Panel B excludes $\ln(GDP)$ and includes lagged $MilShare$. F-stat is the first-stage joint significance test. Eq. Test (p) is the p-value for the difference between peacetime and wartime coefficients. Strong instruments (F > 10) are in bold.

The implementation of gravity-based strategy produces the results reported in Table 12. Crucially, we obtain robust instruments for both export[6] (with the exception of comprehensive export measures) and import sanctions. The estimates confirm a significant causal impact of aggregate export sanctions — an effect likely driven by partial restrictions — on reducing absolute military expenditure. Conversely, no statistically significant causal effects are identified across the remaining specifications.

[6]The quality of the instrument for any export sanctions is approaching the threshold of significance, enabling me to interpret these results as well.

# 6 Discussion

## 6.1 Purging the Endogeneity Bias

Reflecting on the baseline OLS results, one might conclude that most types of sanctions stimulate military growth. However, once we account for endogeneity, the causal estimates reveal that no sanction type exerts a significant impact on military expenditure during peacetime. This confirms the presence of a strong selection bias and provides clear support for **Hypothesis 1**: countries targeted by sanctions are often those that have already initiated military expansion for internal reasons. OLS misinterprets this pre-existing trend as a response to external pressure. Consequently, sanctions appear to function more as a marker of international disapproval regarding a state's "bad behavior" rather than a driver of defense policy, neither triggering a rally-round-the-flag effect strong enough to alter defense budgets nor acting as an effective signal for appeasement by incentivizing less confrontational fiscal choices.

## 6.2 Heterogeneity by Sanction Type

Our disaggregated analysis provides a more granular view of these dynamics. The most straightforward category — prohibitions on the supply and sale of weaponry (arms sanctions) — shows no causal effect on spending. This likely points to successful substitution strategies, where target states secure equipment from domestic producers or non-aligned third parties. While this forced transition to sub-optimal alternatives may degrade actual military capabilities (due to inferior technical specifications or higher cost-to-quality ratios) it does not compel the target to adjust its aggregate military outlays in monetary terms.

In contrast, military assistance sanctions — which include restrictions on monetary aid, technical training, and logistical support — exert a clear negative impact, but *exclusively during wartime*, in turn, referring us to **Hypothesis 2**, which finds confirmation here. In a high-intensity conflict, the blockage of spare parts and specialized logistics creates a physical bottleneck that is far harder to circumvent than the procurement of finished weapon systems. Furthermore, the withdrawal of financial aid specifically earmarked for defense limits a state's capacity to fund its own protection — a constraint that is less binding in peacetime when reliance on external military subsidies is generally lower. In this scenario, the target state may possess the political will to spend but lacks the actual means to execute those outlays.

The effect of trade sanctions is generally negligible but becomes significant for certain sub-types, specifically partial (sectoral) restrictions on exports from sender states (which constitute import restrictions for the target). Crucially, this effect is absent in peacetime, likely due to the target's greater room for maneuver and ability to find alternative trading partners. During wartime, however, a military economy already operating at capacity faces severe difficulties when specific supply chains are severed, leading to an involuntary reduction in planned defense spending.

Interestingly, this effect disappears when sanctions transition into a comprehensive export ban. we argue that total isolation triggers a full-scale resource mobilization within the target state, prompting it to maintain or even increase military outlays despite a shrinking economy. This is supported by our findings that under complete sanctions in peacetime, military expenditure increases as a share of GDP even if absolute spending does not, signaling a shift toward a "siege economy", serving as a clear illustration of the correctness of **Hypothesis 3**.

Finally, no identifiable causal impact was found for other categories. For financial and travel sanctions, the data failed to yield a sufficiently robust instrument, likely because these "smarter", more targeted measures are less predictable based on macro-political factors. Diplomatic and formal restrictions ("other" sanctions) are unique in that they show no significant correlation with spending even in OLS models. This underscores their role as expressive tools of basic diplomatic grievance, utilized in contexts far removed from the intentions of inducing a shift in a target's core military capabilities.

## 6.3 Methodological Contribution: The Role of Gravity-Based Weights

A significant portion of our empirical success stems from the application of gravity-based trade weights. By weighting sanction sensitivity according to exogenous shares of bilateral trade, we obtained significantly more reliable instruments than those provided by simple counts or GDP-weighted measures. Economic vulnerability is fundamentally dyadic and specific; it is not a linear function of a sender's aggregate GDP nor equal across all partners. Ignoring these structural realities increases the dispersion of estimates and obscures the impact of coercive measures. The proposed gravity weights allow us to isolate the "true" structural dependence of a recipient on a sender, effectively purging the measure of political noise and contemporaneous shocks.

## 6.4 Summary of Causal Effects

Table 13 provides a concise summary of the causal impacts identified across the various sanction specifications.

Table 13: Summary of 2SLS Causal Findings by Sanction Type

| Sanction Type | Effect in Peacetime | Effect in Wartime |
|---|---|---|
| **Aggregate Sanctions** | No impact | No impact |
| **Arms Embargoes** | No impact | No impact |
| **Military Assistance** | No impact | Negative |
| **Export Sanctions (Target's Imports):** | | |
| *Partial Restrictions* | No impact; decrease in GDP share | Negative |
| *Complete Restrictions* | No impact; increase in GDP share | No impact |
| **Import Sanctions (Target's Exports):** | | |
| *Partial Restrictions* | No impact | No impact |
| *Complete Restrictions* | Slightly positive | No impact |
| **Financial Sanctions** | No impact | No impact |
| **Travel Sanctions** | No impact | No impact |
| **Other (Diplomatic/Formal)** | No impact | No impact |

## 6.5 Policy Implications

The empirical findings of this study offer several strategic insights for the design and implementation of international coercive measures. First, the discrepancy between OLS and 2SLS results suggests that the perceived effectiveness of aggregate sanctions in peacetime is largely illusory. For policymakers, this implies that broad, multi-country coalitions do not automatically translate into a reduction of a target's military capabilities or a shift in its budgetary priorities. In a peaceful context, regimes appear highly capable of absorbing the political signal of sanctions without adjusting their long-term military investment. Consequently, aggregate sanctions in peacetime should be viewed primarily as tools of diplomatic expression or normative signaling rather than as instrumental mechanisms for curbing militarization.

Second, our results highlight a critical distinction between arms embargoes and military assistance sanctions. While prohibitions on finished weapon systems (arms sanctions) show no causal impact on spending (likely due to the ease of market substitution), the restriction of military assistance (logistics, technical support, and aid) proves highly effective during active conflicts. This suggests that the most impactful check on a warring state's military machine is the creation of physical and financial bottlenecks in its supply chain. For sanctioning states, targeting the specialized logistical and maintenance inputs of a target's existing arsenal provides more leverage than attempting to block the acquisition of new, high-profile platforms.

Third, the findings regarding trade sanctions warn against the potential counterproductive effects of comprehensive embargoes. We find that while partial export restrictions (import constraints for the target) can lead to an involuntary reduction in absolute military spending during wartime, transitioning to a complete trade ban may trigger a "siege economy" response. In such cases, the target regime prioritizes the survival of the defense sector above all else, resulting in an increase in the military's share of a shrinking GDP. From a policy perspective, this indicates that calibrated, sectoral pressure is more likely to constrain a target's resource allocation than total economic isolation, which tends to consolidate the regime's control over remaining resources.

Finally, the success of the gravity-based weighting strategy emphasizes that the economic vulnerability of a target is dyad-specific. Sanctioning coalitions should be assembled based on the structural trade dependence of the target rather than the aggregate GDP of the senders. A sanction imposed by a major global economy with negligible trade ties to the target is effectively symbolic; effective coercion requires the participation of partners that sit at the target's specific geographic or historical trade nodes. Purging the political noise from these trade shares allows for a more accurate identification of where the target's budget is truly vulnerable to external shocks.

# 7 Conclusion

This study investigated the causal relationship between international economic sanctions and the defense allocations of target states. By moving beyond observational estimates, which suffer from severe endogeneity and selection bias, this paper developed a modificated 2SLS framework. The incorporation of gravity-predicted bilateral trade weights allowed for a precise, exogenous measurement of target vulnerability, purging the analysis of contemporaneous political shocks and bilateral trade distortions.

The empirical analysis establishes three primary conclusions. First, the widely observed positive correlation between sanctions and military expenditure is largely an artifact of reverse causality. In peacetime, exogenous sanction shocks do not induce a statistically significant alteration in a target's absolute military outlays or its military burden. Aggregate sanctions, therefore, function primarily as tools of normative signaling and diplomatic expression rather than effective mechanisms for forced demilitarization.

Second, the efficacy of coercion is highly conditional on both the target's conflict status and the specific nature of the restriction. While arms embargoes and financial restrictions demonstrate no robust causal impact, targeted military assistance sanctions and partial export restrictions force a significant contraction in defense spending exclusively during wartime. This supports a physical bottleneck mechanism: in a war economy operating at full capacity, the severance of specific logistical and supply chains deprives the state of the material ability to execute defense outlays (regardless of the regime's political intent).

Third, the results caution against the deployment of comprehensive trade embargoes. The data suggest that transitioning from partial restrictions to total economic isolation can trigger a "siege economy" response. Faced with a shrinking GDP, target regimes tend to protect or even increase the defense sector's share of the national budget, consolidating resources to ensure regime survival and military readiness.

These findings carry direct implications for the design of coercive statecraft. Policymakers seeking to degrade a hostile state's war-making capacity should prioritize targeted logistical and sectoral bottlenecks over broad, multi-country aggregate embargoes. Furthermore, sanctioning coalitions must be assembled based on the specific structural trade dependence of the target, as the economic weight of a sender is irrelevant if it lacks established, gravity-driven trade linkages with the recipient.

While the gravity-based IV strategy addresses the primary endogeneity concerns, several limitations remain. The reliance on reported macroeconomic figures and external military expenditure estimates (primarily those provided by SIPRI) precludes the observation of illicit trade networks, sanctions evasion, and the informal military economy. Although we do not question a methodology employed by SIPRI to reconstruct defense outlays even for opaque regimes, these figures remain estimates that may not fully capture off-budget military funding or clandestine resource reallocations. Furthermore, due to the high collinearity between specific sanction types, isolating the independent effect of simultaneous trade restrictions remains econometrically challenging.

A significant caveat also concerns the use of military expenditure as a proxy for actual military capability. Defense outlays measure the financial resources allocated to the sector, but they do not necessarily reflect changes in military capabilities or technological sophistication, particularly under sanction pressure. Coercive measures, especially arms embargoes and dual-use technology bans, often inflate the unit cost of procurement by forcing target states to utilize clandestine supply chains or invest in sub-optimal domestic substitutes. Consequently, a sanctioned regime might maintain a stable military budget while experiencing a sharp decline in its real war-making capacity, as the purchasing power of its defense currency is eroded by higher transaction costs and the adoption of inferior technologies.

Future research should aim to disaggregate military expenditure itself, examining whether sanctions cause an internal reallocation between personnel costs, equipment procurement, and research and development. Additionally, exploring how different authoritarian institutional structures (e.g., personalist versus military juntas) condition the budgetary response to exogenous sanction shocks would further refine our understanding of economic coercion.

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# 8 Appendix

As a further robustness check, Table 14 presents an alternative specification that more closely replicates the zero-stage methodology of (Kwon et al., 2022). By utilizing the substantial sample size of aggregate sanctions (rather than disaggregated types), we are able to employ a probit model at the zero-stage without encountering the "perfect separation" problem. As shown, the resulting coefficients and standard errors deviate only marginally from those reported in the primary specifications (Table 8). Notably, however, the instrument strength in this probit-based approach is lower than that achieved through the LPM used in our main analysis.

Table 14: Robustness Check: 2SLS Estimations with Probit Zero-Stage (Aggregate Sanctions)

| | **DV:** $\ln(MilExp_{it})$ **(Absolute)** | | **DV:** $MilShare_{it}$ **(Share of GDP)** | |
|---|---|---|---|---|
| | **(1)** | **(2)** | **(3)** | **(4)** |
| **Variables** | Log Count | Log GDP-Wgt | Log Count | Log GDP-Wgt |
| Sanctions (Peacetime) | 0.040 | 0.006 | 0.002 | 0.000 |
| | (0.038) | (0.006) | (0.003) | (0.000) |
| Sanctions (Wartime) | -0.194 | -0.026 | -0.005 | -0.001 |
| | (0.166) | (0.022) | (0.017) | (0.003) |
| War (Main Effect) | 0.108 | 0.143 | 0.011* | 0.013 |
| | (0.083) | (0.111) | (0.006) | (0.010) |
| $\ln(GDP_{it})$ | 0.072*** | 0.070*** | | |
| | (0.015) | (0.014) | | |
| Polity | 0.001 | 0.001 | -0.000* | -0.000 |
| | (0.001) | (0.001) | (0.000) | (0.000) |
| Polity$^2$ | -0.000 | -0.000 | 0.000 | 0.000 |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| Lagged Dependent Var. | 0.873*** | 0.873*** | 0.654*** | 0.660*** |
| | (0.022) | (0.021) | (0.230) | (0.226) |
| **Fixed Effects** | | | | |
| Country FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| **IV Diagnostics & Stats** | | | | |
| First-Stage F-Statistic | **22.23** | **39.73** | **20.38** | **32.20** |
| Equality Test ($p$-value)$^\dagger$ | 0.205 | 0.181 | 0.697 | 0.724 |
| Observations | 4,918 | 4,918 | 4,965 | 4,965 |
| $R^2$ (uncentered) | 0.919 | 0.919 | 0.484 | 0.484 |

*Notes:* Robust standard errors (HC1) in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.10$. Models are estimated using 2SLS on demeaned data. Unlike the main specifications, the instrument here is derived from a zero-stage **Probit model** (rather than a Linear Probability Model).

$^\dagger$ The Equality Test reports the $p$-value for the $F$-test of the null hypothesis that the effect of sanctions in peacetime equals the effect in wartime ($H_0 : \gamma_{peace} = \gamma_{war}$).

Table 15: Gravity Model Estimates (PPML)

| **Variables** | **(1)** **Target's Exports** ($trade\_flow_{ji}$) | **(2)** **Target's Imports** ($trade\_flow_{ij}$) |
|---|---|---|
| $\ln(GDP_{sender})$ | 0.855*** | 0.714*** |
| | (0.000) | (0.000) |
| $\ln(GDP_{target})$ | 0.902*** | 0.840*** |
| | (0.000) | (0.000) |
| $\ln(Distance_{ij})$ | -0.611*** | -0.587*** |
| | (0.000) | (0.000) |
| Contiguity | 0.257*** | 0.537*** |
| | (0.000) | (0.000) |
| Common Language | -0.117*** | 0.006*** |
| | (0.000) | (0.000) |
| Colonial Link | 0.437*** | 0.437*** |
| | (0.000) | (0.001) |
| Intercept | -35.208*** | -29.930*** |
| | (0.003) | (0.003) |
| Observations | 194,892 | 194,728 |
| Pseudo $R^2$ | 1.000 | 1.000 |

*Notes:* Standard errors in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.1$. The model is estimated using Poisson Pseudo-Maximum Likelihood. The dependent variable is the dyadic trade flow in current US dollars. Column (1) estimates the export flow from the target country $j$ to sender $i$; Column (2) estimates the import flow to target $j$ from sender $i$.

Table 15 presents the estimates from the PPML gravity models used to generate exogenous bilateral trade weights. As established in the trade literature (Silva and Tenreyro, 2006), PPML is the preferred estimator for capturing trade flows due to its robust handling of heteroskedasticity and zero-trade observations. The coefficients align with the standard gravity framework: economic mass (GDP) of both partners positively and significantly drives trade, while geographic distance acts as a substantial friction. These fitted values allow us to calculate trade shares that are determined strictly by geography and economic size, ensuring the exogeneity of our weighting scheme.

Table 16 reports the results of the zero-stage LPM, which predicts the likelihood of sanction imposition at the dyadic level. Across all primary categories (aggregate, military assistance, and trade sanctions) excluded instruments demonstrate high statistical significance ($p < 0.01$). Specifically, the historical sanctioning aggressiveness of the sender and the lagged political misalignment (UNGA voting distance) are strong predictors of current sanctioning behavior. The inclusion of target-country and year fixed effects ensures that we account for time-invariant country-specific vulnerabilities and global temporal shocks. These predictors and their performance at the dyadic level provide the empirical basis for the high first-stage F-statistics observed in main country-year specifications.

Table 16: Zero-Stage Linear Probability Model (LPM) Estimates

| **Variables** | **(1) Aggregate** | **(2) Military** | **(3) Trade** |
|---|---|---|---|
| Lagged Log Aggressiveness | 0.011*** | 0.002*** | 0.010*** |
| | (0.000) | (0.000) | (0.000) |
| Lagged UNGA Distance | 0.023*** | 0.000*** | 0.051*** |
| | (0.003) | (0.000) | (0.001) |
| Interaction (Aggr $\times$ Dist) | 0.038*** | 0.010*** | -0.009*** |
| | (0.001) | (0.001) | (0.001) |
| $\ln(Distance_{ij})$ | -0.002*** | 0.000*** | -0.010*** |
| | (0.000) | (0.000) | (0.000) |
| Contiguity | 0.023*** | -0.000*** | -0.011*** |
| | (0.002) | (0.000) | (0.001) |
| Common Language | -0.000 | 0.000*** | -0.011*** |
| | (0.001) | (0.000) | (0.000) |
| Colonial Link | -0.011*** | 0.012*** | -0.010*** |
| | (0.002) | (0.000) | (0.001) |
| Intercept | 0.020*** | -0.000*** | 0.086*** |
| | (0.002) | (0.000) | (0.001) |
| Fixed Effects | Year, Target | Year, Target | Year, Target |
| Observations | 884,178 | 884,178 | 884,178 |
| $R^2$ | 0.0171 | 0.0177 | 0.0128 |

*Notes:* Robust standard errors in parentheses. $***p < 0.01$, $**p < 0.05$, $*p < 0.1$. The zero-stage is estimated at the dyadic level (sender-target-year) using a Linear Probability Model. All specifications include year and target-country fixed effects. The dependent variable is a binary indicator for the imposition of the respective sanction type.